
skip: closed-book answer
VAD end of turn
user speech
x_1
x_2
x_3
x_4
x_K
Streaming ASR + VAD
belief b_k
remaining speech $\hat{R}_k$
Controller (see below)
Retriever L
LLM
TTS
time to first audio (TTFA)
(a) Joint (Ours, Ours-S)
at each partial: dispatch now or wait, using b_k and $\hat{R}_k$ together
dispatch time depends on the belief
(b) Decomposed (B9, B9+)
timing rule (no belief)
retrieve iff $U(b) > 0$ at trigger
no shared state
(c) Speculative (B10, B11)
prefetch early (B11: belief gate)
use at end of turn iff $U(b_K) > 0$
an unused prefetch costs compute only